

Course Overview

Welcome to the course "**Advanced Neural Network Techniques**". This course is a part of the specialization "**Foundations of Neural Networks**". This course dives into more complex architectures and methodologies in neural networks. Students will learn about advanced models that can handle sequential data, generative tasks, and deep reinforcement learning applications. The focus is on practical implementations and innovations in deep learning, enabling students to tackle sophisticated problems.

Further, this course is structured using 4 modules:

- **Module 1:** Recurrent Neural Networks
- **Module 2:** Auto Encoders
- **Module 3:** Generative Deep Neural Networks
- **Module 4:** Deep Reinforcement Learning

Module 1: Recurrent Neural Networks- Covers the architecture and application of recurrent neural networks (RNNs) for sequence data, including time series and natural language processing. Students will explore how RNNs maintain context and memory, which is crucial for tasks like speech recognition and text generation.

Module 2: Auto Encoders- Explores autoencoders as a tool for unsupervised learning, discussing their structure, training methods, and use cases such as dimensionality reduction and anomaly detection. Students will learn how autoencoders can be utilized for feature extraction and data compression.

Module 3: Generative Deep Neural Networks- Examines generative models, including GANs (Generative Adversarial Networks), discussing their theory, architecture, and real-world applications. Students will engage in hands-on projects to create their own generative models and explore ethical considerations surrounding their use.

Module 4: Deep Reinforcement Learning- Introduces the principles of reinforcement learning, focusing on deep learning techniques that enable agents to learn through interaction with their environment. Students will understand the trade-offs between exploration and exploitation, and apply algorithms to solve complex decision-making tasks.

In this course, you will explore advanced models that can handle sequential data, generative tasks, and deep reinforcement learning applications.

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✓ Completed

